

Cybersecurity Analysis of Noon (KSA)

CYS 403

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Introduction

The importance of digital technologies has changed the everyday activities of people including shopping, communication, and financial transactions. In these activities, the role of electronic commerce has considerably increased and allows users flexible and easy options to buy from and sell to people from all over the world. The role of electronic commerce has been considerably magnified in the Kingdom of Saudi Arabia. Among the many activities in KSA, the role of electronic commerce is enabling the fulfillment of Vision 2030, the KSA long-term strategy to achieve economic and digital activities diversification.

Noon is a good example of a Riyadh-based company actively contributing to the improvement of electronic commerce in KSA. Established in 2017, Noon has grown into a comprehensive digital ecosystem offering a wide range of services including online retail, grocery delivery, logistics, and digital payments. Noon has a regional market, offering services in KSA, the United Arab Emirates, and Egypt. Noon serves millions of people and manages a large number of transactions every day.

The increase of electronic commerce in KSA and reliance on Noon has to change the priorities of electronic commerce. The protection of Noon's system greatly depends on risk control. cyber systems used in electronic commerce and cyber systems used to manage and control these systems. The protection of the privacy of users and the maintenance of trust and the continued support of the cyber system of electronic commerce is of the utmost importance.

In addition, the nature of e-commerce platforms presents unique security difficulties. Unlike traditional systems, platforms like Noon include multiple participants, like customers, third-party sellers, logistics providers, and payment systems. Users of the system, including customers, sellers, logistics providers and payment contractors, all pose security risks. Thus, it is important to analyze security gaps and derive ways to overcome them.

Executive Summary

The intent of this document is to provide a report outlining a full range of up-to-date issues on cyber-attacks and security targets focused on Noon, online commerce in the Kingdom of Saudi Arabia (KSA). Noon is quickly becoming an online commerce industry leader and is involved in various services from online retail, online commerce logistics, and online digital payments. Noon started in 2017 and has quickly focused on creating digital economic solutions to contribute to one of the fastest growing digital economies globally, the KSA and wider Middle East.

The larger the platform, the more the attack surface. Digital commerce services (like Noon) deal on and offline with transactional, behavioral, and sensitive user information. This information, and the server in which it is housed, are gold mines to phishing schemes, account takeovers, payment fraud, and data breaches. Defense against attacks is becoming a growing concern as Noon expanded third party sellers and services in order to make the platform more valuable.

This report assesses Noon by conducting a survey of what security measures are in place. Noon, according to the platforms privacy and security policy statements, has parental oversight of user data and systemic measures of encryption and firewalls. Despite the measures, risks of third-party sellers and social engineering do present the greatest concern.

This report outlines threats and assesses their probability and consequences. For classification, the risk is segmented based on hardware, software, data, people, procedures, networks, and sub-components around information systems. Control mechanisms are proposed, such as stronger authentication, enhanced seller verification, and enhanced monitoring systems.

Also, the report outlines the existing Noon security policy and guidelines and assesses the protection and coverage provided to the users and the level of compliance to the accepted and recognized standards. An improved security policy is proposed based on this assessment, including multi-layered security, artificial intelligence-powered fraud detection, and monitoring and assessment of suspicious actions.

Company/System Overview

This study will focus on Noon. Noon is one of the largest e-commerce platforms in the Middle East. Noon is based in Riyadh in Saudi Arabia. Noon was founded in 2017. Noon was founded as part of the digital transformation and economic diversification campaign in Saudi Arabia as part of Vision 2030. Noon has major strategic backing from the Saudi Public Investment Fund and other major regional investments.

Noon is more than an online store. Noon runs an entire digital ecosystem. Noon's primary platform is a marketplace for a wide range of consumer goods. These goods include electronics, clothes, appliances, and even groceries. Noon has gapped marketplace services and integrated grocery and food delivery, logistics, and digital payment services.

Noon has its own payment solution as part of its digital ecosystem. It is called Noon pay. Noon pay allows customers to process all transactions securely while remaining within the Noon platform. Noon Pay increases user convenience. It, however, brings about some potential critical considerations for windowed systems regarding how online user payment data will remain secure.

Noon serves and connects a wide range of stakeholders. Noon connects customers, third-party sellers, payment service providers, and logistics and fulfillment services. Noon operates a multi-sided marketplace to maximize service and product offerings. However, this makes architectures system complex.

Noon uses a combination of web-based applications and mobile applications to provide services. As users access services via web and mobile applications, backend services are utilized to manage product catalogs, process orders and payments, and coordinate deliveries. The system

also includes warehouses and fulfillment centers that assist with supply chain operations to optimize storage and fulfill orders.

The system of Noon establishes and retains several different types of data such as personally identifiable information (name, address, and phone number) of users, information pertaining to financial transactions (order and payment), and information of users who sell. This data is vital to the functioning of the system, and the platform is exposed to a significant risk if there is a data breach. For example, if shell vendor accounts were to be managed for fraudulent purposes, the platform would be exposed to a greater risk of potential trust.

There are also specific risks with vendor shell accounts. The model aims to increase the product and service catalogue, and potential growth, but increases the risk of fraudulent products and services, and supply chain data.

For the digital marketplace of Saudi Arabia, Noon is a big and complex example of a digital information system that aims to integrate technology, and for the digital marketplace of Saudi Arabia, it is important because of its role of study for cybersecurity. However, for the size and complexity of Noon's use case, it loses the digital marketplace for Saudi Arabia, thus, increasing the need for a close look at the security.

Current Security Situation Analysis

The security situation of Noon photography combines strong technical and operational security, but also has obvious nagging weaknesses as a result of the nature of the electronic commerce framework. Noon, as a major electronic commerce giant in the region from Saudi Arabia and with coverage across the Middle East, has to deal with the intricacies of securing a marketplace that contains users, sellers, payment systems, and logistics.

Existing Security Measures

Noon employs a number of protective measures to keep user information secure and to ensure the company's business line functions safely. To secure payment information, Noon, like many companies today, employs the security practice of HTTPS to protect the user who visits the website to protect him from using the information to log in to him and to protect him from using his payment data.

Also, Noon has several payment monitoring systems that also serve as payment screening systems to detect payment fraud. Noon employs several transaction fraud detection controls and several payment transactions screening and system risk controls to improve the efficiency and reliability of payment fraud controls.

In Saudi Arabia, Noon prohibits sellers from manipulating reviews or selling counterfeits, and from violating both the payment and fraud control policies. Noon conforms to these fraud

policies to the letter. To protect its users and its business line, Noon strenuously enforces this policy.

Last but not the least, the platform provides user awareness guidance to reduce user risks. To Noon, the human factor is as equally part of the security environment as the technical component.

Identified Security Issues

These systems provide some level of protection, but many problems persist because of the distributed and open characteristics of e-commerce systems.

1. Phishing and Social Engineering Attacks

Noon customers are subjected to phishing, where attackers pose as sending parties to collect credentials and sensitive information. Noon has active disclaimers informing users to keep passwords and OTPs confidential and to be careful of fake messages and websites.

The effectiveness of phishing naturally stems from the human factor.

2. Third-Party Seller Risks

Noon provides a marketplace for third-party sellers to place products for selling. Although this is positive for variety, it brings the following risks:

- The sale of counterfeit and low-quality products
- Seller fraud
- Changes in the manipulation of ratings and reviews

Noon is known to place strict laws and punishments, but these may be ineffectively implemented in the larger scope.

3. Account Hijacking and Unauthorized Access

Accounts can be taken over for many reasons, such as poor password strength, password reuse, and phishing.

- Once access is gained, attackers can:
- Make purchases
- Take advantage of payment methods already set up in the profile
- Retrieve personal data

A sizable number of users have reported unauthorized purchase cases and/or odd behavior, pointing to the immediate need for more rigorous monitoring.

4. Payment System Limitations and Failures

Noon has powerful systems to detect almost all forms of fraud, but still, there can be payment related issues such as:

- Blocked transactions
- Authorization issues with certain cards
- Countless delays or an intermingled malfunction in the payment system

Although designed with a focus on protecting users, the system has more than its share of issues.

5. System Availability and Service Disruptions

Noon's payment system may sometimes undergo periods of maintenance such that there may be delays in the processing of transactions. There may even be instances of disruptions of the processing of payments. This may be due to scheduled maintenance of the payment system or unexpected issues with one of the gateway systems.

While delays of this nature may be inevitable in matters of system security resulting from periodic maintenance, high availability is a necessary consideration in the operation of any system.

Human Factor and User Behavior Risks

The security risk contribution level of the whole system of Noon shows that most of the risk comes from user behavior and not from system weaknesses. The major user behavior risks are:

- Weaknesses in user passwords whereby some users are likely to choose easy passwords and some use the same password for multiple services.
- Falling victim to phishing.
- The challenge of communicating protected information through email and other unprotected systems.

Noon users are notified that no sensitive information such as their password and OTP are communicated through applications such as WhatsApp or Telegram. This indicates phishing of such nature.

Overall Security Assessment

Overall, Noon has a moderately positive security posture from:

- Encryption, secure communications.
- Detection of fraud systems.
- Policies and regulation of sellers.
- Activities related to user awareness.

Despite all these, Noon retains significant risk exposures, especially in:

- The management of third-party seller risks.
- Phishing and Social engineering.
- Irregular user authentication.
- System unreliability.

These are not Noon-centric problems, but challenges of all large-scale e-commerce platforms. However, these should be a priority for maintaining users and their trust so as to provide sustained and secure services.

Risk Management Plan

One of the most critical aspects of cybersecurity is effective risk management. This is even more important for large scale e-commerce platforms like Noon. The number of cyber threats targeting online platforms across Saudi Arabia is on the rise. Organizations attempting to remain on the right side of the law and employ best practices for risk management in these situations are forced to adopt more sophisticated and structured methods of risk identification, assessment, and management. The National Cybersecurity Authority (NCA) have also stressed the need for a comprehensive national cybersecurity framework.

In the case of Noon, the need for risk management arises due to the complexity of the ecosystem, comprised of numerous customers, third party sellers, payment providers, and support service providers. The following sections are a short synopsis of the most important aspects of the risk management process.

Risk Identification

The most important aspect of risk management is identifying the threats to the system, and the primary components of the e-commerce system are fluid and continue to change based on the rate of expansion of the digital economy and the proliferation of online sales.

For Noon, the most significant risks include:

- **Data breaches**
Accessing customer data, including their name, email, offline address, telephone number, and payment information, without consent.
- **Account takeover attacks**
Attacks that help breach and compromise a user account, including credential harvesting.
- **Payment fraud**
Utilizing breached payment information in order to facilitate payments to a compromised account, or payments to a third party (usually a seller).
- **Third party Seller fraud**
Selling counterfeits and misrepresenting the actual products offered to customers.
- **Phishing and social engineering**

Using compromised information to defraud the user of information, including log on and passwords.

- **System downtime / service disruption**

Failures that affect service availability.

- **Insider Threats**

In this case, breach of process occurs when the platform employees and partners that have privileged access (that is, are cleared) misuse contractual privilege.

The outlined risks are typical of the type of cybersecurity threats present in Saudi Arabia's digital economy that relates to the rapid proliferation of connectivity in the economy.

Risk Assessment and Evaluation

Risks should be evaluated on two key components:

- **Likelihood (Probability of occurrence)**
- **Impact (Severity of consequences)**

In Saudi Arabia, the most common sophisticated cyber-attacks are phishing and ransomware attacks. For companies operating in Saudi Arabia, these attacks are of the utmost priority.

Example Risk Evaluation

Risk	Likelihood	Impact	Overall Risk
Phishing attacks	High	High	Critical
Account hijacking	High	High	Critical
Data breach	Medium	Very High	Critical
Seller fraud	High	Medium	High
System downtime	Medium	High	High
Insider threat	Low	High	Medium

Risk Classification (Six Information System Components)

To deal with risks systematically, they can be classified into the six components of Information Systems (IS):

1. Hardware

Which can include servers, data centers and warehouse systems (Risks: hardware failure, physical damage)

2. Software

Which can include the frontend and the backend of systems (Risks: vulnerabilities, bugs, malware)

3. Data

Most systems are data-centric and risks include data leaks and data breaches (Risks: data leaks, unauthorized access)

4. People

Users, employees, and sellers which poses risks of human error and phishing attacks

5. Procedures

Which can include weak policies and poor enforcement (Risks: weak policies, poor enforcement)

6. Networks

Which can encompass (Risks: interception, DDoS attacks)

This classification allows one to deal with the risks of a system from all the components of an IS

Risk Control Techniques

To deal with risks, an organization should take preventive, protective, and corrective actions. Such actions would be in accordance with the Saudi cybersecurity frameworks and practices.

Technical Controls

- Encryption of sensitive data (HTTPS, TLS)
- Shields and systems to detect and defend against attacks
- Secure applications
- Regular assessments of vulnerabilities

Administrative Controls

- Strong and sustained protective systems
- Consistent and active training and education of employees
- Verifying and scrutinizing employees and sellers

Operational Controls

- Continuous and consecutive monitoring of employees
- Rapid response to emergencies
- Systems to recover the data

Preventive vs Detective Controls

Type	Example
Preventive	Multi-factor authentication
Detective	Fraud detection systems
Corrective	Incident response & recovery

Alignment with Saudi Cybersecurity Frameworks

Saudi Arabia has provided considerable frameworks and guidance in cybersecurity through the National Cybersecurity Authority (NCA) in relation to the risk of the components across sectors.

These frameworks approach the following aspects:

- Identifying and assessing risk.
- Ongoing surveillance.
- Responding to incidents.
- Adhering to local regulation.

Regulations like the Personal Data Protection Law (PDPL) also necessitate organizations to deploy stringent measures to control the storage and processing of customer data, particularly in a cloud environment.

Authentication and Data Confidentiality Evaluation

When it comes to cybersecurity, authentication and data confidentiality are crucial in the e-commerce space, and Noon is no exception. Noon ensures only authorized individuals can access the system, and the sensitive data is protected from exposure. Since Noon's services consist of personal, payment, and transactional data, the protection and confidentiality of that data are crucial to the services offered to customers, and so is the protection of the data and services.

Authentication Mechanisms

Noon uses several authentication methods that are an effective and convenient balance between ease of use and security.

1. Username and Password Authentication

To an extent, some of the most popular and widely used authentication methods involve using either an email or a phone number (often referred to as the username) in conjunction with a password. These methods are prevalent given the ease of use and implementation. A counterargument for the prevalence of this method is that this method is affected by attacks like brute force, credential stuffing, and password reuse.

Noon portrays the need for a strong password to prevent a breach and promotes a 'no password sharing' policy to reinforce the need for users to comply with these secure practices.

2. One-Time Password (OTP) Verification

To provide a supplemental degree of security, Noon uses One-Time Passwords (OTP) in an SMS format, for login and transactional integrity. The 'one-time' nature of these credentials means that they provide a temporary, albeit lasting, integrity and security of the system.

There remains a gap between security and assurance on every authentication method using an electronic medium. Although, in some cases, the OTP authentication methods provide a reasonable degree of assurance against the implications of stolen passwords, there do exist a host of cases in which the assurance can be entirely obliterated due to the means of the attack.

3. Session Management

To secure user sessions, Noon uses several session management techniques such as:

- Session expiration after inactivity
- Secure handling of login sessions
- Logout mechanisms to prevent unauthorized reuse

These features control the risks of session hijacking, the primary aim of which is taking over an existing user session.

4. Current Authentication Limitations

Ongoing concerns include

- Forced mandatory accounts requiring MFA.
- Over-reliance on SMS OTP.
- Password hygiene concerns.

These concerns address the fact that Noon considers the level of authentication security to be relatively low, and improves upon the control mechanisms.

Data Confidentiality Mechanisms

Data confidentiality is the practice of ensuring sensitive data is restricted to a defined set of storage locations and transport mechanisms, and is protected from unauthorized access and use.

1. Encryption in Transit

Noon secures communications via the Internet with payment card specifics and confidential information via HTTPS and other protected mechanisms. For example, data is requested via the exchange of secure means.

With information systems, Noon applies payment data control via their systems.

2. Encryption and Data Protection Measures

Noon maintains the protection of confidential information via the application of advanced information systems that defend the information systems via their protection mechanisms. Noon employs the systems of the:

Noon implements application systems to manage controls over internal resources on behalf of third parties. For example, via information systems, Noon uses a privileged role to control the internal resources of.

5. Payment Data Security

An advantage of Noon's strategy is the decision not to store payment card specifics directly on their servers. This initiates the protection of confidential payment systems from the improper storage of more than a single piece of large-scale payment card specifics.

This method follows top industry standards for the safeguarding of financial data.

6. Access Control Mechanisms

In the system, access to data is limited and defined by one's specific role and responsibilities. This ensures:

- Sensitive data is exposed to only those who need to know
- Data exposure is kept to the absolute minimum
- Internal misuse is kept to the absolute minimum.

These access controls are particularly relevant in systems of large scale and complexity where multiple parties are involved in accessing the system.

Effectiveness Assessment

On the whole, Noon's authentication and data privacy methods incorporate a mid-range security option, ensuring basic encryption and OTP as standard industry security practices.

Strengths

- Encrypted client-to-server communications (HTTPS)
- Secure Payment Processing (no card storage)
- Role-Based Access Controls
- OTP-based verification

Weaknesses

- No mandatory MFA
- Reliance on SMS Authenticated
- Phishing and social engineering risk
- User awareness and responsibility

Security Policies and Guidelines Analysis

Security policies and guidelines are implemented to instruct systems users on best practices in security. For an e-commerce platform like Noon, security policies are necessary to safeguard users' data, control seller actions, and adhere to legal and cybersecurity frameworks within Saudi Arabia.

This section will examine Noon's security policies and determine privacy policies, obligations, and procedural guidelines to security to determine their effectiveness in maintaining security.

Overview of Noon's Policies on Security

Noon has several policies and guidelines detailing the handling of user data concerning maintaining security on the platform. Among these, the protection of personal data collection, processing, and storage, as well as personal data collection and processing, is of utmost importance.

Noon's policy guarantees the protection of user data. Noon will ensure that there is no unauthorized access, no disclosure of evidence, and no evidence of ministerial evidence alteration.

Noon's data security control policy will ensure the protection of user data through the following:

- Keeping data and systems secure through encryption during transmission
- Employing the use of firewalls to protect the systems
- Restricting the access of data only to authorized personnel

Furthermore, Noon assures its customers that when processing payment data, it will comply with industry standards concerning the security and safety of the processing and collection of payment data.

User Security Guidelines

Noon helps user account security with personal data tips. These tips include:

- Keep your password to yourself.
- Don't share your one-time passwords.
- Always log out on shared devices.
- Look out for phishing attempts.

Noon's service signs say users are responsible for protecting their passwords. Users are instructed to take action against their accounts being accessed without authority.

Noon shows other contents urging users to keep watch against their account being compromised, meaning they acknowledge threats against their account being hacked.

Data Handling and Privacy Compliance

Noon's policies explain the handling of personal data, including:

- User data collection (name, contact info, payment info)
- Data storage and processing across systems associated with Noon
- Data use for transactions, services, and support

Noon also mentions that due to being a global platform, data movement and processing across systems is common, which raises security concerns.

Also, Noon is a signatory to laws and regulations, including the Protection of Personal Data Law (PDPL) in Saudi Arabia.

Seller Policies and Marketplace Guidelines

Noon, as a marketplace, has policies to regulate third-party sellers, such as:

- Engaging in fraudulent and deceptive behavior is banned
- Restrictions on the sale of counterfeit and fake products
- Product descriptions must be truthful and accurate
- Imposition of suspensions and/or fines

These policies help manage risks toward third-party sellers and help build a framework of trust on the platform. However, because of the volume of sellers, it may be difficult to enforce some of these policies.

Evaluation of Policy Effectiveness

Noon has enacted a multitude of policies. Still, the guidelines are to be seen as effective both with enforcement and adherence.

Strengths:

- Transparent policies on data and confidentiality
- Indicator of data protection and privacy level among industry peers (PCI DSS)
- Policies articulate; user obligations
- Policies to regulate the marketplace and reduce fraud

Weaknesses:

1. Over-Reliance on User Responsibility

Numerous guidelines place the onus on the user (e.g. password protection & phishing avoidance). This is problematic as users are usually lacking understanding of guidelines & laws user responsibility creation of phishing loopholes, etc.

2. Limited Transparency in Enforcement

Enforcement is subjective at best. For example, there are regular reports of problematic counterfeit products & seller policies.

As demonstrated in the community discussions, there are more legal violations when sellers operate as third-party sellers compared to when sellers operate as regular sellers.

3. No Guarantee of Complete Security

Noon claims there are no guarantees that all information sent via the Internet is protected, which is a cyber policy disadvantage & exaggeration of accuracy.

4. Data Sharing Across Systems

The transfer of data between the systems and the sharing of data between systems and regions creates a potential exposure to structured & unregulated security folds.

Alignment with Security Standards

Noon's policies align with:

- Industry best practices like (encryption & access)
- The structured policies & regulations of payment (PCI DSS)
- Data protection principles (confidentiality & controlled access)

However, there is limited evidence of policies & standards since the adoption of maker policy (layered) access standards for the privileged (PPP). It is believed that there are best practices for no fences policy & more integrated, multi-purpose fences.

Proposed Enhanced Security Approach

The current security analysis of Noon has validated the existence of several security posture weaknesses characterized by poor authentication mechanisms, third party seller risks, weakness to Phishing and Social Engineering attacks, and several others. Noon has done well to implement a baseline set of security postures, but like every other organization, Noon is not insulated from the fast-developing nature of risks.

In consideration to recent improvements, this section will analyze a possible security solution modeled to the contemporary security practices and proactive implementations which impact positive security of the platform

1. Implementation of Mandatory Multi-Factor Authentication (MFA)

One of the most basic implementations is providing and/or imposing Multi-Factor Authentication (MFA) especially when a user is logging into the platform and also when financial transactions are being made.

Presently, Noon is relying heavily on Passwords and OTPs which are sent through SMS services. OTPs, being a second factor, is believed to add a layer of security to the authentication process. This Too is vulnerable, and in the recent years, attacks like SIM swapping, or phishing attacks have also become popular in circumventing this form of authentication. Through the years, the vulnerabilities of the MFA implementations done by Noon at this point are several. Therefore, suggestions on better MFA implementations include,

- Biometric (fingerprint, facial recognition, or both)
- App-based (Authenticator Apps)
- Device-based (Trusted Devices)

In the Modern Cybersecurity Sphere, MFA, which has also been referred to as 2FA, is one of the primary deterrents of unauthorized access and reducing attacks to a great extent

2. Adoption of Zero Trust Architecture (ZTA)

A Key suggestion to Noon is to have a Zero Trust Architecture (ZTA) as a reference point for its operations.

One of the principles of Zero Trust is, never trust, always verify. This means that whether a user or a system is situated within or within the organization's periphery, they should not be trusted by any means.

Important Features to Implement:

- Continuous authentication and verification
- Access controls and least-privilege principles
- System micro-segmentation
- Activities real-time logging and system alerts

The primary advantage of the defensive architecture is the isolation of system components. This both prevents unauthorized access and limits the impact of actual breaches.

3. AI-Based Fraud Detection and Behavioral Analytics

In light of the evolving cyber threat landscape, Noon is recommended to incorporate AI and ML into the protection architecture.

Behavioral AI systems analyze users to identify behavioral anomalies caused by:

- Unusual login locations
- Abnormal transaction activity
- Unexpected account activity

Such systems can autonomously execute counter-measures such as:

- Transaction blocking

- Increased authentication
- Security team notifications

AI systems address the contemporary cyber threat landscape as they are capable of perpetual detection and response.

4. Seller Verification and Seller Monitoring

Because third party sellers are the main source of risk, we have to implement the following:

- Sellers may have to pass a KYC check
- Sellers may have to show ID to be verified
- Sellers' activities will be monitored
- Fraudulent listings will be detected automatically

Sellers may have a score to indicate their reputation to inform users on purchase decisions.

5. User Protection and Security Education

Many of the attacks to secure systems rely on the ignorance of the user and therefore greater user education is critical.

Examples of user education may include:

- Alerts to users to inform them of suspicious system actions
- Messages to users to inform them of suspected system fraud
- Instructions to users of proper and secure system use
- Instructions to users on proper and secure system use

Cyber systems are weak, and a true layer of defense is to inform greater control systems users.

CIA Triad Analysis

The CIA is comprised of three security components: confidentiality, information integrity, and system availability. This means data is kept from being accessed by unauthorized users, data is not tampered with, and data is accessible when needed.

In the case with Noon, the CIA triad is being employed to incorporate a security framework for analysis of current and future system security measures.

Confidentiality

Confidentiality is the prevention of infringing upon a user's space and private information. This means users private data such as their address, payment, and purchase history is omitted from prying eyes.

Current Status in Noon

For the integrations of confidentiality, Noon utilizes:

- Transportation and storage of data in HTTPS
- Imposed restrictions to data
- Non-accessible payment integrations

However, phishing/scamming attacks, poor/insecure passwords and lack of a newly advanced authentication Level create security breaches.

Improvements from Proposed Approach

The proposed security enhancements confidentiality by improving and tightening the level of security from:

- Multi-Factor Authentication (MFA) → blocks account(s) from breaches.
- Biometric authentication → easy integration and further protection for breaches.
- More advanced Zero Trust Architectures → blocks a user from breaches.
- Improvements in encryption and tokenization → sets security perimeters for data in storage and for to/from data breaches.

Impact

The aforementioned improvements relieve stress of breaches, hijacked accounts, and high security oceanic levels of vault breaches.

Collectively, these improvements further enhance the users privacy, safety, integrity of data, and reliability on the platform.

Integrity

Integrity preserves the consistency, accuracy and trustworthiness of data. When data corruption is at risk, integrity provides the shield against data corruption and unauthorized changes during storage or transmission.

Current State in Noon

Noon preserves integrity of data through:

- Containment of system access
- Processing of secure transactions
- Validation of data on the back end

The integrity of Noon's system is not sustainable due to:

- Fraud involved by certain sellers
- Product listings that have been tampered with
- User accounts that have been modified

Improvements from Proposed Approach

The system will also improve integrity of Noon by:

- The development of an AI fraud detection system to find and intervene in active fraud situations
- The factual verification of the identity of sellers to better control who is selling on the platform
- Improving the accuracy of financial transactions through validation systems
- Comprehensive functionality of the system will be monitored through logs and audits with the intent to identify anomalies

Impact

With these systems in place, Noon can improve the integrity and credibility of the platform through:

- Real-time data checks to ensure that the integrity of the data is reliable
- Protection from unauthorized data changes

Availability

Availability ensures the integrity of systems and data and gives an assurance to stakeholders during peak periods of demand.

Current State in Noon

Noon provides a high degree of availability with:

- Infinite scalability of the infrastructure
- The platform remains fully operational
- Ongoing maintenance of the site and its monitoring systems

Challenges:

- Periodic disruptions of services
- Occasional outages of the payment gateway
- High traffic during peak periods

With the planned changes, Noon will improve availability in the following ways:

- The redundancy of systems and backup infrastructure
- The improvement of the assurance of systems against interruptions and of control systems in place
- Reduced risk of downtime through better designs of the data center

Impact

With these improvements, interim customers will face:

- Permanent construction disruptions
- Robust system infrastructure
- Reduced chaos during peak periods.
- Integrating the systems will be a good way to improve and sustain the business continuously.

Balance Between CIA Components

An important issue facing the CIA triad is the enforcement of balance. This means there is a potential for one aspect to affect the others negatively. Some of the possible examples of this include:

- Complete confidentiality controls may impede system ease of use.
- Maximal availability with no control may increase susceptibility.

Thus, it may be said that such organizations with a triad of CIA should work to relax their controls to implement easy operational use with a system of comprehensive controls over all factors.

Overall Evaluation

Based on the analysis:

CIA Component	Current Level	After Improvements
Confidentiality	Moderate	High
Integrity	Moderate	High
Availability	Moderate–High	High

Conclusion

To conclude, we discussed Noon's approach towards dealing with cyber security as an e-commerce platform in Saudi Arabia. Fundamental security measures like encryption, authentication, etc., are implemented, but risks associated with phishing, user behavior, and interactions with third-party sellers are still present.

In the report, we suggest implementing a structured risk management approach, which would include multi-factor authentication, AI-based monitoring (to address risks associated with monitoring), and extension of security fundamentals to the Zero Trust Architecture. This added focus on the extension to the CIA triangle of the security fundamentals (i.e., protecting data from

unauthorized disclosure and accuracy, as well as data availability, constantly) would address the proposed security deficiencies.

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